§2 Axiom

公理体系：假设、定义、公理.

I. 量子态 ∈ Hilbert空间，线性.

① 态矢 $c|\psi\rangle + d|\phi\rangle$

② 右矢 $c^*\langle\psi| + d^*\langle\phi|$

③ 内积 $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$

④ 基矢展开 $\langle i|j\rangle = \delta_{ij}$, $\sum_i |i\rangle\langle i| = 1$

$$|\psi\rangle = \sum_i c_i |i\rangle, \quad c_i = \langle i|\psi\rangle$$

$$\langle\psi| = \sum_i c_i^* \langle i|, \quad c_i^* = \langle\psi|i\rangle$$

连续化：

$$\langle x|y\rangle = \delta(x-y), \quad \int dx\, |x\rangle\langle x| = 1$$

$$|\psi\rangle = \int dx\, \psi(x)|x\rangle, \quad \psi(x) = \langle x|\psi\rangle$$

$$\psi(x) = \langle x|\psi\rangle = \int dy\, \langle x|y\rangle\langle y|\psi\rangle$$

$$= \int dy\, \delta(x-y)\,\psi(y)$$

⑤ 基矢变换

$$\begin{cases} \langle a_i|\psi\rangle = \sum_j \langle a_i|b_j\rangle\langle b_j|\psi\rangle \\ \langle\xi|\psi\rangle = \int d\eta\, \langle\xi|\eta\rangle\langle\eta|\psi\rangle \end{cases}$$

eg. Fourier变换

$$\psi(x) = \int dk\, \langle x|k\rangle\langle k|\psi\rangle = \int dk\, \frac{e^{ikx}}{\sqrt{2\pi}}\,\psi(k)$$

eg. 连续 ↔ 离散基矢变换

$$\psi(x) = \sum_n \langle x|n\rangle\langle n|\psi\rangle = \sum_n f_n(x)\, c_n$$

$$\langle n|\psi\rangle = \int dx\, \langle n|x\rangle\langle x|\psi\rangle = \int dx\, f_n^*(x)\,\psi(x)$$

$|n\rangle$ 可以是能量本征态.

II. 可观测量作为厄米算符.

① $A|\psi\rangle = |\phi\rangle$

② 展开：

$$A = \sum_{ij} |i\rangle\langle i|A|j\rangle\langle j| = \sum_{ij} |i\rangle A_{ij}\langle j|$$

$$|\psi\rangle = A|\psi\rangle = \sum_{ij} |i\rangle A_{ij}\psi_j \Rightarrow \psi_i = A_{ij}\psi_j$$

$$AB = \sum_{ijk} |i\rangle A_{ij} B_{jk}\langle k| \Rightarrow (AB)_{ik} = A_{ij}B_{jk}$$

③ 期望值. $\langle\psi|A|\psi\rangle = \sum_{ij} \psi_i^* A_{ij}\psi_j$

④ 本征值、本征态.

$A|a\rangle = a|a\rangle \Rightarrow |a\rangle$ 是 A 本征态，对应本征值为 a.

对于分量 i：$A|a_i\rangle = a_i|a_i\rangle$

⑤ 伴随算符

$$\langle\psi|A\phi\rangle = \langle A^\dagger\psi|\phi\rangle = \langle\phi|A^\dagger\psi\rangle^*$$

$$\Rightarrow A^\dagger_{ij} = A^*_{ji}$$

⑥ 厄米算符 $A^\dagger = A$

$a_i\ a_i = a_i^*$：

$$a_i = \langle a_i|A|a_i\rangle = \langle a_i|A^\dagger|a_i\rangle^* = \langle a_i|A|a_i\rangle^* = a_i$$

$$\langle a_i|A|a_j\rangle = a_j\langle a_i|a_j\rangle = \langle a_i|A^\dagger|a_j\rangle = a_i^*\langle a_i|a_j\rangle = a_i\langle a_i|a_j\rangle$$

$$\Rightarrow \langle a_i|a_j\rangle = \delta_{ij}$$

② 投影算符 $P = |a\rangle\langle a|$

③ 幺正变换 $U: U^\dagger U = 1 \Rightarrow \langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle$

$\langle\psi'|A|\psi'\rangle = \langle\psi|U^\dagger A U|\psi\rangle$

view: $|\psi\rangle \to U|\psi\rangle$ / $A \to U^\dagger A U$

表象变换是一种幺正变换

$|a\rangle \to |b\rangle$, $U = \langle b|a\rangle$

无穷小幺正变换 $U: |\psi\rangle \to U_\varepsilon|\psi\rangle$, $U_0 = I$

Taylor展开: $U_\varepsilon = U_0 + \varepsilon U_0' + \cdots = I + i\varepsilon G$

幺正性: $U_\varepsilon^\dagger U_\varepsilon = (I - i\varepsilon G^\dagger)(I + i\varepsilon G)$

$= I + i\varepsilon(G - G^\dagger) + O(\varepsilon^2) = I$

$\Rightarrow G = G^\dagger$

有限变换由无穷小生成元: $U_\varepsilon = e^{i\varepsilon G}$

$U_\varepsilon^\dagger A U_\varepsilon = (I - i\varepsilon G^\dagger) A (I + i\varepsilon G)$

$= A + i\varepsilon(AG - G^\dagger A) = A + i\varepsilon[A, G]$

若 $[A, G] = 0 \Rightarrow AU = UA$

四、量子化条件

1. 经典力学中：Poisson 括号

$$\{A,B\} = \frac{\partial A}{\partial q}\frac{\partial B}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial B}{\partial q}$$

→ Hamilton 方程 $\begin{cases} \dot q = \{q, H\} \\ \dot p = \{p, H\} \end{cases}$ * $\frac{\partial q}{\partial q} = \frac{\partial p}{\partial p} = 0$

相空间中物理量 $A(q,p)$

$$dA = \frac{\partial A}{\partial q}dq + \frac{\partial A}{\partial p}dp = \left(\frac{\partial A}{\partial q}\frac{dq}{dt} + \frac{\partial A}{\partial p}\frac{dp}{dt}\right)dt$$

$$= \left(\frac{\partial A}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial H}{\partial q}\right)dt = \{A,H\}\,dt$$

→ 什么时间平移的生成元

对于平移变换 $T(a): (q,p) \to (q+a, p)$

$$T(a)\,A(q,p) = A(q+a,p)$$

$$= A(q,p) + \frac{\partial A}{\partial q}a + O(a^2)$$

$$dA = \frac{\partial A}{\partial q}\cdot da = \{A,p\}\,da$$

→ 广义平移变换的无穷小生成元

量子力学中：$|\psi\rangle \in H$，$\hat U(\varepsilon)$ 将 $|\psi\rangle$ 平移 ε

Taylor 展开：$\hat U(\varepsilon) = \hat U(0) + \left.\frac{d\hat U(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0}\varepsilon + O(\varepsilon^2)$

$$= I + \left.\frac{\partial \hat U(\varepsilon)}{\partial \varepsilon}\right|_0 \varepsilon = I + \varepsilon\hat K$$

平移算符幺正 $U^+U = (I+\varepsilon\hat K^+)(I+\varepsilon\hat K) = I$

→ $\varepsilon(\hat K^+ + \hat K) = 0$，$\hat K$ 是反厄米算符。

对位置本征态 $|x\rangle$ $\left(\hat x|x\rangle = x|x\rangle\right)$

$$\hat U(a)|x\rangle = |x+a\rangle$$

$$\hat x\,\hat U(a)|x\rangle = (x+a)|x+a\rangle$$

$$\hat U^+(a)\hat x\,\hat U(a)|x\rangle = (x+a)|x\rangle$$

→ $\hat U^+(a)\,\hat x\,\hat U(a) = \hat x + a\hat I$

经典（对易

设 $\hat{U}(a) = e^{-\frac{i}{\hbar}\hat{p}a}$，$\hat{p}$ 为动量算符

$$e^{\frac{i}{\hbar}\hat{p}a}\hat{x}e^{-\frac{i}{\hbar}\hat{p}a} = \hat{x} + a$$

$$(1+\frac{i}{\hbar}\hat{p}a)\hat{x}[1-\frac{i}{\hbar}\hat{p}a] = \hat{x} + a$$

$$[\hat{p},\hat{x}] = -i\hbar$$

$$\Rightarrow [\hat{q},\hat{p}] = i\hbar \quad 其中 \quad [a,b] = ab - ba$$

动量算符 $\hat{p} = -i\hbar\frac{\partial}{\partial x}$ 在坐标表象下的形式

$\psi(\vec{r}) = \langle \vec{r}|\psi\rangle$，$\langle \vec{r}|\vec{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i}{\hbar}\vec{p}\cdot\vec{r}}$，$\langle \vec{p}|\psi\rangle = \phi(\vec{p})$，则 $\psi(\vec{r}) = \int \langle \vec{r}|\vec{p}\rangle\langle \vec{p}|\psi\rangle d^3p$

2. Cauchy - Schwartz 不等式：

$$\langle\alpha|\alpha\rangle\langle\beta|\beta\rangle \geq |\langle\alpha|\beta\rangle|^2$$

取 $|\alpha\rangle = |\psi\rangle$，$|\beta\rangle = |\varphi\rangle$ 则

$$\langle\psi|\psi\rangle\langle\varphi|\varphi\rangle \geq |\langle\psi|\varphi\rangle|^2$$

$$= |\langle \{\hat{Q}_\psi, \hat{Q}_\varphi\} + [\hat{Q}_\psi, \hat{Q}_\varphi]\rangle|^2$$

$$\geq |\langle [\hat{Q}_\psi, \hat{Q}_\varphi]\rangle|^2$$

3. 动量算符的本征态和本征值

$$\langle \psi|\hat{x}|\chi\rangle = \langle \psi|\hat{x}\hat{I}|\chi\rangle = \int \langle\psi|x\rangle\langle x|\hat{p}|p\rangle\langle p|\chi\rangle dx dp$$

$$= \int \psi^*(x)\, x\, \chi(x) dx$$

$$\langle \psi|\hat{p}|\chi\rangle = \int \langle\psi|x\rangle\langle x|\hat{p}|p\rangle\langle p|\chi\rangle dx dp$$

$$= \int \psi^*(x)\, p\, \chi(x) dx$$

$$= \int \psi^*(x)\, (-i\hbar\frac{\partial}{\partial x})\, \chi(x) dx$$

$$= \int \chi(x)\, (i\hbar\frac{\partial}{\partial x})\, \psi^*(x) dx$$

$$= \langle \chi|\hat{p}|\psi\rangle^*$$

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}$$

$$\langle x'|\hat{p}|\eta\rangle = -i\hbar\frac{\partial}{\partial x}\langle x'|\eta\rangle$$

$$\hat{Q}|p\rangle = p|p\rangle$$

$$\langle q|\hat{Q} = \langle q|q$$

Fourier Trans:

$$\langle q|\hat{p}|x\rangle = \frac{q}{i\hbar}\langle q|\hat{Q}|x\rangle = \frac{q}{i\hbar}\langle q|x\rangle$$

$$\langle q|x\rangle = C e^{\frac{i}{\hbar}qx} \quad 系数$$

$$\langle q|q'\rangle = \delta(q-q') \quad 归一化$$

$$\delta(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\, e^{ikx}$$

$$= |C|^2 \int dx\, e^{\frac{i}{\hbar}(p-p')x} = |C|^2 \frac{1}{2\pi\hbar}\int dx\, e^{\frac{i}{\hbar}x(p-p')} = |C|^2\, 2\pi\hbar\, \delta(p-p') \Rightarrow C = \frac{1}{\sqrt{2\pi\hbar}}$$

4. 角度量表象下的角动量算符.

坐标表象: $\hat{L}_z = -i\hbar\frac{\partial}{\partial\varphi}$

本征态: $\hat{L}_z|\psi\rangle = L_z|\psi\rangle \Rightarrow \psi(\theta) = C\cdot e^{\frac{i}{\hbar}L_z\theta}$

周期性: $\psi(\theta+2\pi) = \psi(\theta) \Rightarrow L_z = m\hbar$

$\hat{L}_z|m\rangle = m\hbar|m\rangle$, $\langle\theta|m\rangle = \frac{1}{\sqrt{2\pi}}e^{im\theta}$

II. 运动方程

① Schrodinger 绘景.

$|\psi_S\rangle = [1+\frac{i}{\hbar}G(t)]|\psi\rangle = |\psi\rangle + \frac{i}{\hbar}G|\psi\rangle$

表现 $\Rightarrow G = -\frac{H}{\hbar} \Rightarrow H|\psi\rangle = i\hbar\frac{\partial}{\partial t}|\psi\rangle$

② 演化算符.

$$|\psi(t)\rangle = \hat{U}(t,t')|\psi(t')\rangle = \sum_{t'}^{t} e^{-\frac{i}{\hbar}H(t)t}|\psi(t')\rangle$$

$$H|\psi(t)\rangle = H\hat{U}(t,t')|\psi(t')\rangle = i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = i\hbar\frac{\partial}{\partial t}\left[\hat{U}(t,t')|\psi(t')\rangle\right]$$

$\Rightarrow i\hbar\frac{\partial}{\partial t}\hat{U}(t,t') = H\hat{U}(t,t')$

算符 $A(t)$ 期望值的运动方程.

$$\frac{d}{dt}\langle\psi(t)|A(t)|\psi\rangle$$

$$= \frac{1}{\Delta t}\left[\langle\psi+\Delta\psi(t+\Delta t)|A(t+\Delta t)|\psi+\Delta\psi(t+\Delta t)\rangle - \langle\psi(t)|A(t)|\psi\rangle\right]$$

$$= \frac{1}{\Delta t}\left\{\langle\psi+\Delta\psi|\left[A(t)+\dot{A}(t)\Delta t\right]\left[|\psi\rangle+\dot{|\psi\rangle}\Delta t\right] - \langle\psi|A|\psi\rangle\right\}$$

$$= \frac{1}{\Delta t}\left\{\langle\psi|A|\psi\rangle + \langle\Delta\psi|A|\psi\rangle + \langle\psi|A|\Delta\psi\rangle + \langle\psi|\dot{A}|\psi\rangle\Delta t + \langle\psi|A\dot{|\psi\rangle}\Delta t + \langle\Delta\psi|\dot{|\psi\rangle}\Delta t - \langle\psi|A|\psi\rangle\right\}$$

$$= \frac{1}{\Delta t}\left\{\frac{i}{\hbar}\Delta\theta\left[-\langle\psi|AH|\psi\rangle + \langle\psi|HA|\psi\rangle\right] + \langle\psi|\dot{A}|\psi\rangle\Delta t + \langle\Delta\psi|\psi\rangle\right\}$$

$$= \frac{1}{i\hbar}\langle\psi|[A,H]|\psi\rangle + \langle\psi|\dot{A}|\psi\rangle$$

③ Heisenberg 绘景：算符含时

$$\langle \psi(t)|A(t)|\psi(t)\rangle = \langle \psi|U^\dagger(t,0)A(t)U(t,0)|\psi\rangle$$

$$= \langle \psi|A_H(t)|\psi\rangle$$

$$\frac{d}{dt}A_H(t) = \frac{1}{dt}\left[U^\dagger(t+dt,0)A(t+dt)U(t+dt,0) - U^\dagger(t,0)A(t)U(t,0)\right]$$

$$= \frac{1}{dt}\left\{U^\dagger(t)\left(1+\frac{iHdt}{\hbar}+1\right)\left[A(t)+\frac{\partial A}{\partial t}dt\right]\left(1-\frac{iHdt}{\hbar}\right)U(t) - U^\dagger(t)A(t)U(t)\right\}$$

$$= \frac{1}{dt}\left[-\frac{i}{\hbar}dt\left(HA-AH\right) + U^\dagger(t)\frac{\partial A}{\partial t}U(t)\right]$$

$$= \frac{1}{i\hbar}[A(t),H(t)] + U^\dagger(t)\frac{\partial A}{\partial t}U(t)$$

④ 相互作用绘景：

$$H(t) = H_0 + V(t)$$

$$\langle \psi(t)|A(t)|\psi(t)\rangle = \langle \psi|U_0(0,t)U^\dagger(0,t)A(t)U_0(t,0)U_0^\dagger(t,0)|\psi\rangle$$

$$= \langle \psi_I|A_I(t)|\psi_I\rangle$$

$$\begin{cases} |\psi_I\rangle = U_0^\dagger(0,t)|\psi_S\rangle \\ A_I(t) = U_0^\dagger(0,t)A_S(t)U_0(0,t) \end{cases}$$

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle_I = i\hbar\frac{\partial}{\partial t}\left[U_0^\dagger(0,t)|\psi_S\rangle\right]$$

$$= i\hbar\left[\frac{\partial U_0^\dagger}{\partial t}(0,t)|\psi(t)\rangle_S + U_0^\dagger(0,t)\frac{\partial}{\partial t}|\psi(t)\rangle_S\right]$$

$$= \left[i\hbar\frac{\partial}{\partial t}\left(e^{-\frac{iHt}{\hbar}}\right)+U_0^\dagger H\right]|\psi(t)\rangle_S$$

$$= \left(-U_0^\dagger H_0 + U_0^\dagger H U_0\right)|\psi(t)\rangle_S = U_0^\dagger V(t)|\psi(t)\rangle_S$$

又 $V_I(t) = U_0^\dagger(t)V(t)U_0(t)$

$$\Rightarrow i\hbar\frac{\partial}{\partial t}|\psi\rangle_I = V_I(t)|\psi\rangle_I$$